

SPHERE

- The set of points in space which are at a constant distance from a fixed point is called as sphere. The fixed point is called as centre and constant distance is called as radius of the sphere.
- If radius of a sphere is 1 (unity) then it is called as unit sphere.
- If radius of a sphere is 0 (zero) then it is called as point sphere.
- Equation of the sphere with centre (x_1, y_1, z_1) and radius r is $(x - x_1)^2 + (y - y_1)^2 + (z - z_1)^2 = r^2$.
- Equation of the sphere with centre at origin and radius r is $x^2 + y^2 + z^2 = r^2$.
- General equation of the sphere is $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$ whose centre is $(-u, -v, -w)$ and radius is $\sqrt{u^2 + v^2 + w^2 - d}$.
- General equation of 2nd degree equation $ax^2 + by^2 + cz^2 + 2hxy + 2gxz + 2fyz + 2ux + 2vy + 2wz + d = 0$ represents a sphere if:
 - (a) $a = b = c$.
 - (b) $f = g = h = 0$.
 - (c) $u^2 + v^2 + w^2 - da^2 \geq 0$.
- Equation of sphere having (x_1, y_1, z_1) and (x_2, y_2, z_2) are end point of a diameter is $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) + (z - z_1)(z - z_2) = 0$.
- Equation of sphere concentric with the sphere $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$ is $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + k = 0$.
- Equation of sphere concentric with the sphere $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + d = 0$ and
 - (a) Passes through origin is $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz = 0$.
 - (b) Passes through (x_1, y_1, z_1) is $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz = x_1^2 + y_1^2 + z_1^2 + 2ux_1 + 2vy_1 + 2wz_1$.
 - (c) Having radius r is $x^2 + y^2 + z^2 + 2ux + 2vy + 2wz + (u^2 + v^2 + w^2 - r^2) = 0$.
- Through four non coplanar points, one and only one sphere can passes thought it.
- If the four points are coplanar and no three of them are non collinear then many sphere can pass through them provided they are concyclic.
- If the four points are coplanar and three of them are non collinear then no sphere can pass through them.
